

1.

If the graphs adjacency matrix is $A_G = [a_{ij}]$,

then
$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

Given that the adjacency matrix =

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

And the ordering of vertices is a, b, c, d, draw the corresponding graph.

2.

Given the pseudograph:



Determine the adjacency matrix